Week 5

Case Study 1

Data for Questions 1 to 5

Sensor Direction Scaling

A factory uses two calibrated sensors to monitor vibration and heat on a production line. During a preprocessing audit, engineers examine a linear transformation that scales the two sensor directions differently but does not mix them. The transformation is

$$A = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}.$$

The vectors $u_1 = (1, 0)^\top$ and $u_2 = (0, 1)^\top$ represent two independent sensor directions. A vector is an eigenvector of A if multiplying by A only scales the vector without changing its direction. The team wants to identify which directions keep their orientation after the transformation and how the scale factors appear as eigenvalues.

Question 1: What is Au_1?

A) $(3, 0)^\top$

B) $(0, 3)^\top$

C) $(2, 0)^\top$

D) $(1, 3)^\top$

Correct Answer: A

Explanation: $A(1, 0)^\top = (3, 0)^\top$.

Question 2: What eigenvalue corresponds to u_2?

A) 1

B) 3

C) 2

D) 0

Correct Answer: C

Explanation: $Au_2 = (0, 2)^\top = 2u_2$, so the eigenvalue is 2.

Question 3: Which statement is true about u_1?

A) It is not changed by A for any input scale.

B) It is rotated by 90 degrees after multiplication by A.

C) It becomes the zero vector.

D) It is scaled by a factor of 3.

Correct Answer: D

Explanation: $Au_1 = 3u_1$, so the direction is unchanged and the scale factor is 3.

Question 4: What is the trace of A?

A) 6

B) 5

C) 1

D) 3

Correct Answer: B

Explanation: The trace is the sum of diagonal entries: $3 + 2 = 5$.

Question 5: How are the eigenvalues related to the diagonal entries in this matrix?

A) They are equal to the diagonal entries for this diagonal matrix.

B) They must both be zero for every diagonal matrix in this setting.

C) They are found by adding all entries in each row.

D) They depend only on the off-diagonal entries here.

Correct Answer: A

Explanation: For this diagonal matrix, the eigenvalues are the diagonal entries 3 and 2.

Case Study 2

Data for Questions 6 to 10

Customer Feature Coordinate Review

A data analyst is comparing customers using two centered features: response to discounts and preference for premium products. Instead of using the original axes directly, the analyst represents each customer vector in a new basis that separates common movement from opposite movement across the two features. The basis is

$$b_1 = (1, 1)^\top, \qquad b_2 = (1, -1)^\top.$$

A customer vector is

$$x = (4, 2)^\top.$$

The analyst wants coefficients c_1, c_2 such that

$$x = c_1 b_1 + c_2 b_2.$$

Finding these coefficients shows how the same customer can be described using basis directions rather than raw feature coordinates.

Question 6: Which equation is obtained from the first coordinate?

A) $c_1 - c_2 = 4$

B) $c_1 + c_2 = 4$

C) $2c_1 = 4$

D) $c_2 = 4$

Correct Answer: B

Explanation: The first coordinate of $c_1(1, 1) + c_2(1, -1)$ is $c_1 + c_2$.

Question 7: Which equation is obtained from the second coordinate?

A) $c_1 + c_2 = 2$

B) $c_1 = 2$

C) $c_1 - c_2 = 2$

D) $2c_2 = 2$

Correct Answer: C

Explanation: The second coordinate is $c_1 - c_2$, so $c_1 - c_2 = 2$.

Question 8: What is c_1?

A) 3

B) 1

C) 4

D) 2

Correct Answer: A

Explanation: Adding $c_1 + c_2 = 4$ and $c_1 - c_2 = 2$ gives $2c_1 = 6$, so $c_1 = 3$.

Question 9: What is c_2?

A) 0

B) 1

C) 3

D) 2

Correct Answer: B

Explanation: Since $c_1 + c_2 = 4$ and $c_1 = 3$, $c_2 = 1$.

Question 10: Why do b_1 and b_2 form a basis for $\mathbb{R}^2$?

A) They point in the same direction.

B) They both have zero length.

C) They always produce only positive coordinates.

D) They are linearly independent and span $\mathbb{R}^2$.

Correct Answer: D

Explanation: Two linearly independent vectors in $\mathbb{R}^2$ form a basis.

Case Study 3

Data for Questions 11 to 15

Store Performance Feature Reduction

A retail team measures two centered features for stores: weekly footfall variation and average basket variation. The covariance matrix is

$$\Sigma = \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}.$$

PCA selects directions of maximum variance. For this diagonal covariance matrix, the coordinate axes are principal directions. The team wants to reduce the two-feature store summary to a smaller representation while retaining most of the variation across stores. Because the features have already been centered, the covariance entries describe variation around the average store behavior.

Question 11: What is the variance along the first coordinate direction?

A) 1

B) 2

C) 4

D) 0

Correct Answer: C

Explanation: The first diagonal entry of the covariance matrix is 4.

Question 12: Which direction is the first principal component?

A) The second coordinate direction

B) The direction $(1, 1)^\top$

C) The zero vector direction

D) The first coordinate direction

Correct Answer: D

Explanation: The first coordinate has the largest variance, 4, so it is the first principal component direction.

Question 13: What fraction of total variance is explained by the first principal component?

A) 4/5

B) 1/5

C) 1/4

D) 5/4

Correct Answer: A

Explanation: The total variance is $4 + 1 = 5$, and the first component explains 4.

Question 14: If the data is projected onto the first principal component only, how many dimensions remain?

A) 0

B) 1

C) 2

D) 4

Correct Answer: B

Explanation: Keeping one principal component reduces the representation to one dimension.

Question 15: What is the purpose of centering data before PCA?

A) To make every covariance value negative after centering.

B) To remove all variance from the dataset before PCA.

C) To force all eigenvalues to be equal.

D) To express variation relative to the mean.

Correct Answer: D

Explanation: Centering subtracts the mean so PCA analyzes variation around the mean.

Case Study 4

Data for Questions 16 to 20

Product-User Interaction Compression

A recommendation team studies a small product-user interaction matrix before applying matrix factorization to a larger catalog. The rows can be viewed as products and the columns as user groups, with larger entries indicating stronger interaction. The simplified matrix is

$$M = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}.$$

For this diagonal matrix, an SVD can be written as

$$M = U\Sigma V^\top$$

with $U = I$, $V = I$, and singular values 3 and 1. The team uses this small diagonal example to discuss rank, dominant singular directions, and how keeping only the largest singular value produces a low-rank approximation.

Question 16: What is the largest singular value?

A) 1

B) 2

C) 3

D) 0

Correct Answer: C

Explanation: The singular values listed are 3 and 1, so the largest is 3.

Question 17: What does $V^\top$ represent in the SVD expression?

A) The transpose of the right singular-vector matrix.

B) The loss value of the recommendation model after a full training step.

C) The number of users in every batch.

D) The covariance matrix after centering.

Correct Answer: A

Explanation: In $M = U\Sigma V^\top$, $V^\top$ is the transpose of the right singular vector matrix.

Question 18: What rank-one approximation keeps only the largest singular value?

A) $\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$

B) $\begin{bmatrix} 3 & 0 \\ 0 & 0 \end{bmatrix}$

C) $\begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$

D) $\begin{bmatrix} 0 & 3 \\ 1 & 0 \end{bmatrix}$

Correct Answer: B

Explanation: Keeping only singular value 3 preserves the first diagonal component and drops the second.

Question 19: What is the rank of M?

A) 0

B) 1

C) 2

D) 3

Correct Answer: C

Explanation: Both diagonal entries are nonzero, so the matrix has rank 2.

Question 20: Why is SVD useful for dimensionality reduction?

A) It requires every matrix entry to be exactly zero before the factorization.

B) It removes the need for any matrix factorization.

C) It changes all singular values to negative values.

D) It can keep dominant singular directions and discard smaller ones.

Correct Answer: D

Explanation: Keeping the largest singular values gives a lower-dimensional approximation that preserves dominant structure.